BASQUE CENTER
ON COGNITION, BRAIN
AND LANGUAGE

March 9th 2026

Re: Time-Stamp internal pre-registration TICK: #19942, approved on July 26th, 2023

Prof. Clara Martin, member of the Scientific Committee at the BCBL, hereby confirms that the following pre-registration was presented for evaluation and approved on July 26th 2023 (internal system TICK: #19942). Information related to internal procedures has been obscured. A screenshot of the ticket system is attached to this letter.

Sincerely,

Clara Martin

BCBL Researcher
Speech & Bilingualism Group Leader
www.bcbl.eu/clara-martin
IKERBASQUE Research Professor
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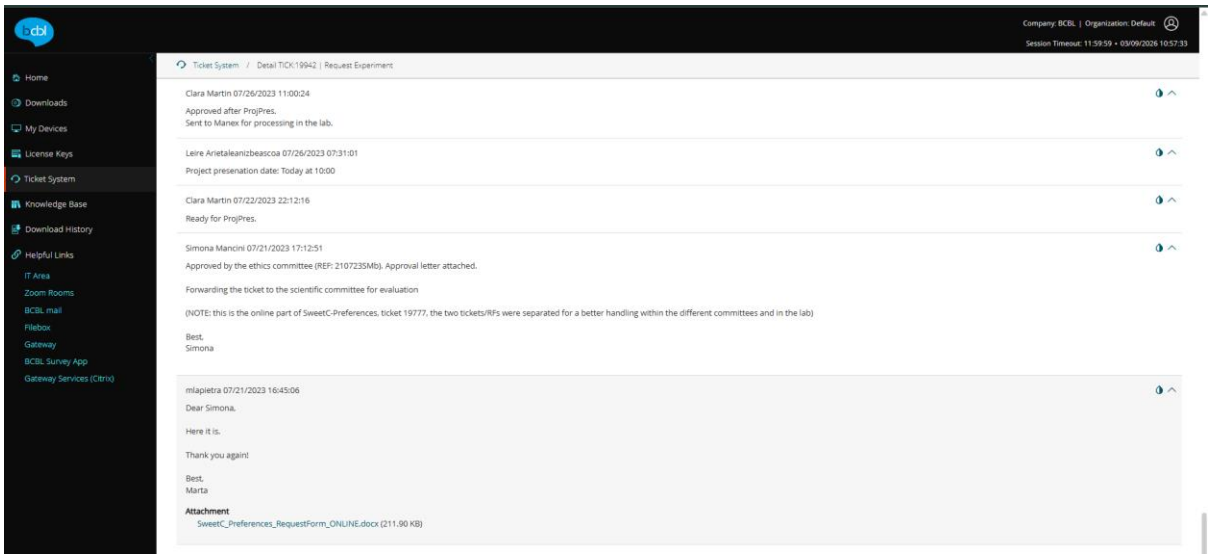

BASQUE CENTER ON COGNITION, BRAIN AND LANGUAGE

Experiment Request Form (ADULTS)

[redacted]

A) BASIC INFORMATION

EXPERIMENT SHORT NAME *(For folder and Participa web)*	SweetC – Preferences (ONLINE)
EXPERIMENT EXTENDED NAME	The "Sweet Spot" of Cognitive Conflict – The Effect of Conflict on Preferences
SUBMITTER	Marta La Pietra
OTHER RESEARCHERS INVOLVED	Manuela Ruzzoli, Marc-Lluís Vives, Nicola Molinaro
DATE OF SUBMISSION	21/07/2023
REQUEST TYPE	**X** Real experiment □ Pilot (more than 5 subjects) □ Normative or survey **X** Follow-up study: Name of previous study: **SWEETC** Ticket Number: **15963**
LOCATION	□ MIRAMON □ KORTA □ MURCIA* **X** ONLINE □ OTHER: _____ □ MULTIPLE LOCATIONS: _____ * For this laboratory, it is necessary to fill in the corresponding annex at the end of this document.
TECHNIQUE	□ Behavioral □ Questionnaires/Survey/Normative* (face-to face) □ Eye-Tracking □ EEG **X** ONLINE (Survey/ normative/experiment) □ MEG □ MRI □ Kinect
SESSIONS	Number of sessions: **1** Duration of sessions: **1h approx.** Session 1: Session 2: Session 3: Session 4: Session 5: Session 6: Total duration of experiment: **1h approx.**

[redacted]

[redacted]

B) EXPERIMENT DESCRIPTION

- Extended description for committees (scientific, ethics, and lab). Give full details of experiment design (aim, hypothesis, script description, methods, data analysis, no word limit). Please explain also why you think this sample size is appropriate (keep in mind there is no single right answer to this question, but try to explain why you believe your sample size is appropriate. If an estimate of the size of an effect is given, specify the unit of measurement too, e.g., Cohen's d).

This is an extension of the already approved SweetC project (TICK:15963; PID2020-114717) in which we aim to test the hypothesis that instances of cognitive conflict can lead to behavioural benefit (i.e., faster reaction times, RTs) or be preferred. In the first experiment, we looked at the behavioural performance in subsequent tasks (e.g., simple RTs, inhibition, memory) after neutral, congruent, and incongruent Stoop trials. A Stage 1 Registered Report of this series of experiments is on its second round of review in the Peer Review Community In RR. This second experiment involves preferences for the level of conflict (i.e., incongruent vs. congruent Stroop trials) and how this choice influences performance and emotional/physical states.

Background

Cognitive conflict is a multifaceted phenomenon described as a mismatch between stimulus and simultaneously activated responses (Botvinick et al., 2001) or as an information gap (Berlyne, 1960). It encompasses violations of expectations (Chetverikov & Kristjánsson, 2016; Cheung et al., 2019; Gruber & Ranganath, 2019), uncertainty (Berlyne, 1957; see also Brandstätter & Herrmann, 2016), surprise (Noordewier & Breugelmans, 2013), errors (Stürmer et al., 2011), and interference in decision-making that guides learning strategies (Abrahamse et al., 2016; Botvinick, 2007). It is an effective trigger for cognitive control functions, flexible behaviour and adaptation (Botvinick, 2007; Botvinick et al., 2001; Dignath et al., 2020).

However, it is considered effortful and aversive (Bouzidi & Gendolla, 2022; Dignath et al., 2020; Dreisbach & Fischer, 2012, 2015; Fritz & Dreisbach, 2013; Inzlicht et al., 2015; van der Wel & van Steenbergen, 2018; Vassena et al., 2017). Its detection is indeed associated with negative affective states, i.e., frustration, feelings of worry, anxiety, caution, and avoidance (Botvinick, 2007; Inzlicht et al., 2015). According to the affective-signalling hypothesis (Dignath et al., 2020), conflict elicits a negative affective response, which triggers control strategies to attenuate the conflict's subsequent impact on performance and reduce aversive reactions. In a relevant study supporting this perspective, Dreisbach & Fischer (2012) asked participants to classify as positive or negative the emotional valence of a word presented after a congruent or incongruent Stroop trial (MacLeod, 1991; Stroop, 1935). Their findings showed that target words with a negative valence were evaluated faster after incongruent than after congruent Stroop trials, and were interpreted as suggesting that incongruency in the Stroop task held a negative valence (see also Fritz & Dreisbach, 2013; Pan et al., 2016).

It is undeniable that cognitive conflict demands effortful control (Bouzidi & Gendolla, 2022; Dignath et al., 2020; Dreisbach & Fischer, 2012, 2015; Fritz & Dreisbach, 2013; Inzlicht et al., 2015; van der Wel & van Steenbergen, 2018; Vassena et al., 2017). However, albeit costly, effort can be stimulating, rewarding and sought-after (Inzlicht et al., 2018; Wu et al., 2022). Indeed, conflict detection and resolution trigger adaptive adjustments in the brain and behaviour, which are ultimately beneficial. Previous studies showed that when participants *resolved* the conflict in a Stroop task and then classified a word's emotional valence

as positive or negative, correct responses on incongruent trials facilitated positive affective reactions compared to congruent trials (Ivanchei et al., 2021; Schouppe et al., 2015).

Within the *Predictive Coding* framework, Clark (2018) has suggested that surprising/unpredictable events – thus, conflict – may be positively interpreted as opportunities for exploration. Clark's (2018) argument is reminiscent of Berlyne (1957, 1960), who believed that a moderate level of conflict could stimulate curiosity and act as a reward. Although Berlyne's idea influenced later works (Botvinick & Cohen, 2014; Gruber & Ranganath, 2019; Huskey et al., 2018; Inzlicht et al., 2018), it has received little experimental support.

Therefore, conflict is negative and aversive for some (Botvinick, 2007; Botvinick et al., 2001; Dignath et al., 2020; Dreisbach & Fischer, 2012, 2015; Fritz & Dreisbach, 2013; Harmon-Jones et al., 2010; Inzlicht et al., 2015; Kerns et al., 2004; Shackman et al., 2011; Shenhav et al., 2013; van der Wel & van Steenbergen, 2018; Vassena et al., 2017; Yeung, 2014) but, at the same time, can also positively impact cognition and behaviour, and being sought after. We detect a contradiction, which can be resolved by shifting perspective and putting more emphasis on the positive value of cognitive conflict.

Hypotheses

In the present work, we put forward the hypothesis that the "right amount" of cognitive conflict can be preferred and sought after by people. To test this hypothesis, we employ a Stroop task (MacLeod, 1991; Stroop, 1935) and ask the participants to freely choose the level of conflict on each block. At the end of each block, participants are asked to rate how much they liked the just-performed block, how effortful it was and the way they feel on an emotional grid. See below for a full description.

Our hypotheses are thus the following:
The *null hypothesis (H0)* posits that there is no significant difference between people's preferences for different levels of cognitive conflict and the evaluation of their experience.
The first *alternative hypothesis (H1)* posits that people prefer an intermediate or a high level of cognitive conflict and evaluate their experience in this range more positively than at a low level of conflict.
Alternatively, the second *alternative hypothesis (H2)* posits that people prefer a low level of conflict and evaluate their experience more positively at that level than at the other two levels, i.e., high and/or intermediate.

Procedure

Before starting the experiment, participants complete an *Emotional Classification Task* (adapted from Heffner & FeldmanHall, 2022) to map emotional words into an Arousal-Pleasantness grid (see below). Then, they will get familiarized with a 4-response keys mapping, with the Stroop task, and with a variable level of conflict. The main test, involving participants' choices and their emotional ratings, starts if no exclusion criteria are met. Finally, two personality trait questionnaire, i.e., the Need for Cognition (Cacioppo et al., 1984) and the Short Boredom Proneness Scale (SBPS; Struk et al., 2017), are presented at the end of the experimental session (see Fig. 1).

The experiment is created and hosted using Gorilla Experiment Builder (www.gorilla.sc; Anwyl-Irvine et al., 2020). Participants will be recruited through *Prolific* (www.prolific.co), a vastly used platform for online experiments.

The Stroop task: The Stroop task (MacLeod, 1991; Stroop, 1935) is a classic cognitive psychology task during which participants are presented with colour words shown in different colours, which may or may not match the meaning of the word. This creates a non-conflicting congruent (i.e., the colour name matches the colour, e.g., blue) and a conflicting incongruent (i.e., the colour name mismatches the colour, e.g., blue) condition. The participant must indicate the colour of the word, ignoring its meaning.

A fixation cross appears in the middle of the screen for 500 ms. Thereafter, the colour word (Arial font, 50 size) replaces it and stays onscreen for 1000 ms. A fixation cross replaces the word until the response or for 1000 ms. Participants are required to provide a response from the word onset till 2000 ms afterwards.

Participants respond by pressing one of four keys on a keyboard. Specifically, the keys Z, X, N and M will define correct responses for the blue [RGB: 0 0 255], red [RGB: 190 0 0], green [RGB: 0 246 0] and yellow [RGB: 250 250 0] colours. Participants learn the colour-response key mapping during a 20-trial practise block (see below).

Practice: Participants initially practise the colour-response key mapping on 20 congruent Stroop trials, during which the set of valid keys is always available to them on the top of the screen. Thereafter, they practice one block of 30 Stroop trials (50% congruent – 50% incongruent). In both practices, they receive coloured feedback (either a correct green sign or a red error cross) for their performance. They can repeat the practice up to 3 times in case they do not achieve an accuracy equal to or greater than 80% and/or do not present a Stroop effect (RTs incongruent trials > RTs congruent trials).

Familiarization: Participants familiarize themselves with two blocks of 10 Stroop trials comprising two different levels of incongruence, i.e., (1) 90% congruent/10% incongruent, (2) 10/90%, receiving feedback (i.e., green/red fixation cross) for their performance to ensure they understand a variable proportion of congruent and incongruent trials within a block. Concerning the Stroop effect, previous research shows that its magnitude is lower if there are more incongruent rather than congruent elements in the task (Schmidt, 2016, 2019; Schmidt & Besner, 2008). Thus, participants are not excluded if they do not show any Stroop effect during this session.

Main Test: It comprises 10 blocks of 30 trials. At the beginning of each block, participants are asked to choose a preferred level of incongruence (i.e., the number of Stroop incongruent trials they want to perform during the upcoming block) using a slider from 0 to 30. The ratio of congruent and incongruent Stroop trials presented in the upcoming task varies depending on the participants' choice.
Personal ratings: After completing the Stroop task, participants provide their responses to two direct statements addressing enjoyment ("I enjoyed doing this task very much.") and effort ("How much effortful was the last block?"), using a 7-point scale. Thereafter, they indicate their subjective rate on a one-shot click on the Arousal-Pleasantness Emotional Grid, keeping in mind where they placed each word during the Emotional Classification Task (see below). Thereafter, they are invited to take a break before another block begins.

Final Questionnaires: Finally, participants complete two personality questionnaires, the Need for Cognition (Cacioppo et al., 1984) and the Short Boredom Proneness Scale (SBPS; Struk et al., 2017).

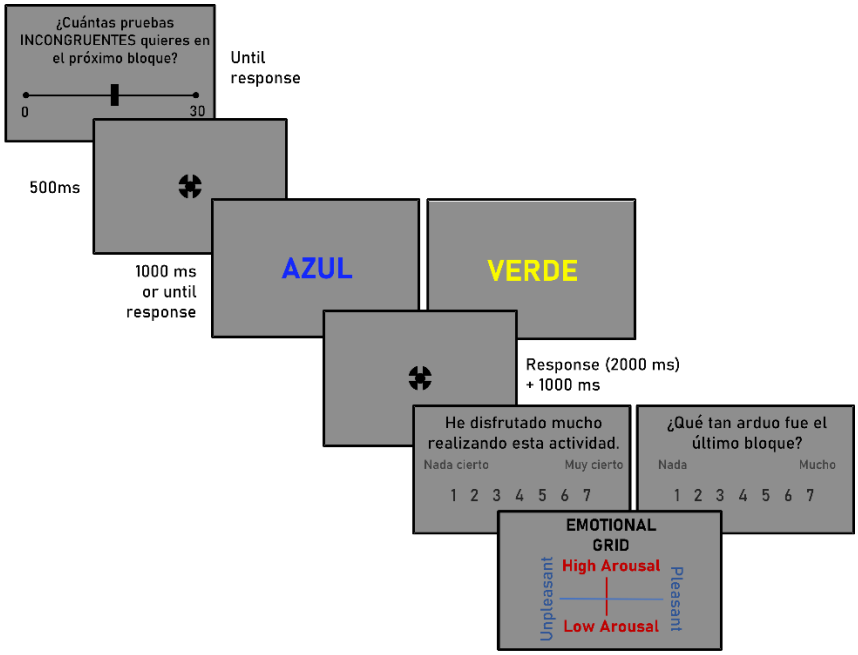

Figure 1. Schematic representation of the paradigm within one block. Participants select the number of incongruent trials they want to perform through the slider at the beginning of each block. After completing a block of the Stroop task with the selected number of incongruent trials, they provide their personal ratings by responding to two statements and by using the Emotional Grid.

Emotional Classification Task: Participants encounter a set of twenty emotion terms, i.e., *afraid, angry, annoyed, aroused, calm, happy, disgusted, enthusiastic, disappointed, nervous, neutral, peppy, quiet, relaxed, sad, satisfied, sleepy, sluggish, still, surprised*. These terms are presented one-by-one above a 500 x 500 pixel grid, in which the centre represents a neutral feeling. The grid has two dimensions: *unpleasant-pleasant* along the x-axis (greater pleasantness corresponding to a farther position to the right), and *high-low arousal* along the y-axis (higher position on the y-axis signifies increased arousal, indicating a heightened state of wakefulness) (Fig. 2). Please note that the arousal dimension is independent on the pleasantness dimension.

Participants' task is to reflect upon what characterized the emotion or state described by the word and click on a position on the grid which they believe best represents the placement of that specific emotion/state along the two dimensions.

At the end of each block, during the main test, participants are presented with the emotional grid only to indicate how they feel after completing the Stroop task.

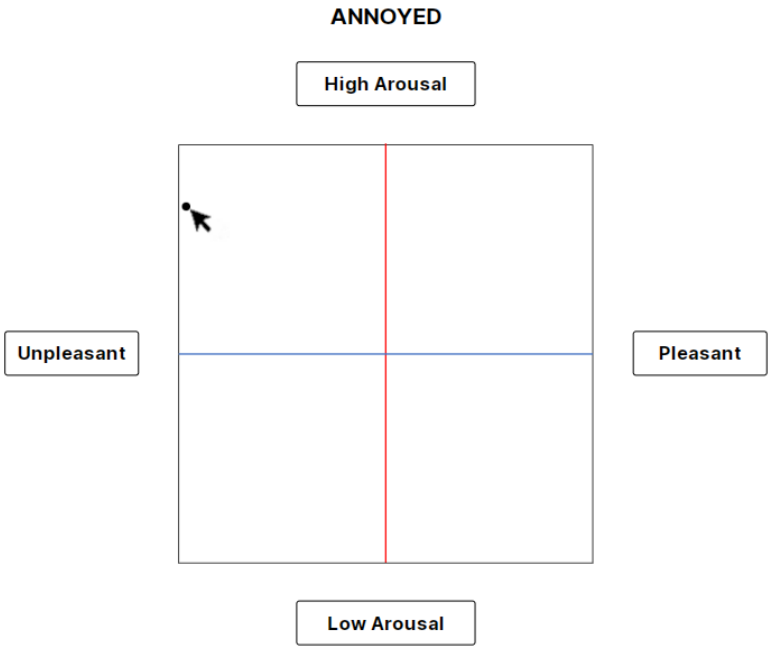

Figure 2. Example of one trial of the Emotional Classification Task. An Arousal-Pleasantness grid is used to create an emotional map for each participant, who needs to click on the grid to subjectively place 20 emotional/state words ("Annoyed" in the example), presented one-by-one.

Questionnaires: At the end of the main test, we will collect data using two questionnaires (attached to the present request) to better characterize the impact of personality traits on results, i.e., the Need for Cognition (Cacioppo et al., 1984) and the Short Boredom Proneness Scale (Farmer & Sundberg, 1986; Isacescu et al., 2017; Struk et al., 2017). Through them, we aim to assess individual tendencies to enjoy effortful cognitive activities and the individual propensity to experience boredom in daily life.

Exclusion criteria

Based on the Stroop performance, some participants might be excluded. The criteria are the following:

Practice

- No Stroop effect: RTs incongruent trials - RTs congruent trials ≤ 0 ms.
- Participants whose response accuracy ≤ 80%.

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C) PARTICIPANT'S REQUIREMENTS

Approximate Age range: (if more than one group needed, please select all)

X Young adults (18-35) □ Special population _____
□ Middle aged (36-60) □ Multi-aged: _____
□ Seniors (>65) □ Other: _____ □ Notes/+info: _____

Languages:

□ Basque-Spanish balanced bilinguals □ Interpreters _____ (please specify)
□ Basque-Spanish unbalanced bilinguals □ Signers _____ (please specify)
□ Almost Spanish monolinguals* (Miramón, Korta) users □ HIKA
□ Almost Spanish monolinguals* (Murcia) **X Other requirements: Spanish native speakers**
□ English native speakers

* Keep in mind that English and Spanish subjects are compulsory at school in Spain, and Basque is present in every school (at different levels depending on the school) in the Basque Country, so there are hardly any Spanish or Basque monolinguals nowadays. If you would like to get more information regarding the linguistic background of the area, please request the lab for help from a linguist RA.

- Please indicate any other relevant information regarding the participant profiles:

Gender: Balanced as much as possible

EXCLUSION CRITERIA:

- **Blind**
- **Colourblind**
- **No vision correction**
- **History of neurological or psychiatric disorders**
- **Substances addiction, abuse, or use**

D) ETHICS FORM AND PARTICIPANTS RECRUITMENT

PARTICIPANT RECRUITMENT

***If you indicated in Section C that your research requires a special population, please include detailed information about participant recruitment in Section L at the end of this form.**

□ BCBL Participa database for the experiment in KORTA
□ Arrangement with an association (specify): _____
□ Your own contacts (specify): _____
X Other (specify): ***Prolific***

[redacted]

ETHICS

- Has the research study been submitted for ethics review elsewhere?
    - **X Yes** □ **No**
    - If so, indicate where and what the outcome was: **TICK:15963, approved by the BCBL.**
- If the participants will be recruited through your own contacts/collaborators, specify whether the participants will be tested at BCBL laboratories or elsewhere. □BCBL □ Elsewhere
    - If you selected 'Elsewhere', specify the location:_____

- If you will receive support from an external organisation for testing your participants, provide full contact details for the organisation and specify whether an agreement has been signed or is in process **PROLIFIC**

- Will the research require recruitment advertising? **X Yes** □ **No**
- Length of experimental session (as reported in section A):
  □ **30 min. or less** □**30-60 min.** **X 60-90 min.** □ **90-120 min.** □ **more than 120 min.**

- For experimental sessions longer than 30 minutes, will participants have breaks at least once every half an hour? **X Yes** □ **No**
- Will the study require filling any questionnaires other than the standard BCBL ones (i.e., language background on Participa, BEST, etc.)? If so, please specify and attach a copy of each: **X Yes** □ **No**
  Need for Cognition, Short Boredom Proneness Scale
- Will the research involve the presentation of emotionally laden material?: □ **Yes** **X No**
    - If so, please upload the stimuli specifying the file names _____

IMPORTANT REMINDER: All consent forms and documents containing personal information of participants must follow BCBL standards in terms of format and content, as well as comply with data protection law policies.

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G) INFORMATION ABOUT THE TASK

Stimuli Type	**X Visual** □ Auditory □ Audiovisual
Response Type	**X Manual** □ Verbal □ Other (specify) _____
Response devices:	□ Touch Pad □ Gamepad □ Response Box □ Headphone □ Microphone **X Other (specify)** **Mouse and keyboard** (online)
Program	□ DMDX □ Matlab □ Presentation □ Experiment Builder □Psychopy **X Other** **Gorilla Experiment Builder** _____

* Please, make a breakdown if it is a multiple-day or a multi-technique experiment

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ONLINE EXPERIMENT SETTINGS:

RECRUITMENT:

□ BCBL Participa database*
□ Your own contact database (specify): _____
□ MTurk*
X Other recruiting platform (specify): **PROLIFIC**

SETTINGS & EQUIPMENT:

□ Browser(s): **Mozilla – Firefox, Microsoft Edge, Google Chrome**
□ Devices in which the experiment works:
X PC □ MAC **X** Laptop
□ Tablet □ Smartphone
□ Other_____

□ Devices participants need to have:

X Mouse **X** Touchpad □ Headphones □ Speakers
□ Integrated microphone □ Detached microphone **X** Keyboard

□ ADDITIONAL REQUESTS/COMMENTS: _____

CATCH TRIALS:
The Ethical Committee has added a compulsory request for online experiments: catch trials. For example, if you are preparing an experiment with auditory stimuli, you will make sure that participants are listening to the audios using some auditory catch trials. This way, we will make sure that the data collection is appropriate.

What kind of catch trials will you use?

Sound

Question about an image

Question about a text

Other : **Require the participants to choose a specific number in a grid**

How many catch trials will appear in your experiment? **8 catch trials**

How many of these will need to be answered correctly by the participant to be accepted to continue with the study? **If participants choose the wrong number in more than 50% (4 catch trials), they will be excluded.**

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J) INFORMATION SHEET

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BASQUE CENTER ON COGNITION, BRAIN AND LANGUAGE

You may request a copy of this sheet to any of our staff members

ADULTS INFORMATION SHEET

SweetC Preferences

Name of the researcher(s) in charge of the project: Dr Manuela Ruzzoli

You have been invited to take part in this research project. Before you decide if you want to participate, it is important that you understand the purpose of this study and what it implies. Please read the following information carefully and do not hesitate to ask us if you need any additional clarification. Thank you very much.

What is our aim with this study?

The SweetC Preferences experiment investigates human's ability to flexibly adapt behaviour to actual environmental situations, using one of the most reliable and known paradigms: the Stroop task, in which a coloured word will be presented on the screen in different ink-coloured, sometimes creating incongruent situations. We believe that sometimes, incongruent situations such as those presented in the Stroop task, could be positive and entertaining. In this study, our goal is to understand what level of incongruence you prefer.

What will you have to do?

Nº of SESSIONS AND DURATION: 1

TOTAL DURATION OF EXPERIMENT: 60 minutes

In this experiment, you will perform a Stroop task. During the Stroop task, you will read several colour words written in several inks. Critically, the ink does not always match the meaning of the word: for example, the word “red” will appear written in red but, in some trials, will appear written in blue/green/yellow. You will need to focus on the colour, inhibiting the meaning of the word, and pressing the corresponding key on a keyboard as fast and as accurately as possible. After practising the task, you will be asked to choose the number of incongruent trials (red written in blue/green/yellow) you would prefer to perform. Once the experiment is over, you will complete some personality questionnaires.

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L) ETHICAL COMMITTEE

RESEARCH INVOLVING ATYPICAL PARTICIPANT POPULATIONS

- **DESCRIPTION OF PARTICIPANT POPULATION: Young adults (18-35 y.o.), no colourblind, no vision impairments (without visual correction), no neurological and psychiatric disorders, Spanish native speakers.**

- **RECRUITMENT OF PARTICIPANTS: (describe the recruitment process in detail)**
  Participants will be recruited through Prolific.

  Will support from an external organisation be needed?

  X No

  ☐ Yes:
  [redacted]

- **PARTICIPANTS' PERSONAL DATA MANAGEMENT:**

(Provide a detailed description of the information that will be used to identify and contact participants and the measures that will be taken for data protection, including, but not limited to participant identifying information, contact details, medical information, diagnosis, etc.)

Participants will proceed with the experiment only after signing the Open Brain consent form and providing their DNI and ProlificID (online).

- **ETHICAL ASPECTS AND RISKS:**

  a) **Does this participant belong to a <u>vulnerable population</u>* (see descriptions below)?**
  X No ☐ Yes (specify)_____

  b) **Will individual participants in this study encounter difficulties to access, read, or understand the full information provided in the informed consent form?** **X No** ☐ Yes (specify)_____

IMPORTANT NOTE: if you responded YES to any of the two previous A and B questions, provide detailed information of the measures taken to mitigate such risks:

_____

_____

*** DESCRIPTIONS OF VULNERABLE POPULATIONS:** (If in doubt, please discuss your case with the BCBL ethics committee):

Cognitive or Communicative Vulnerability: The category of cognitive or communicative vulnerability broadly encompasses persons who have difficulty comprehending information and making decisions about participation.

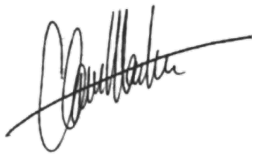
Institutional Vulnerability: The category of institutional vulnerability includes persons who are under the formal authority of others who might have different values, goals, and priorities than those of the potential research participant.

Deferential Vulnerability: Like institutional vulnerability, the deferential vulnerability category includes persons who are under the authority of others.

Medical Vulnerability: The medical vulnerability category includes persons who have serious health conditions for which no satisfactory standard treatment options are available.

Economic Vulnerability: The economic vulnerability category includes persons who are disadvantaged in the distribution of social goods and services such as income, housing, or healthcare.

Social Vulnerability: The social vulnerability category includes persons who belong to undervalued social groups.

(Definitions taken from: http://www.ochsnerjournal.org/content/20/1/34#sec-5)

Clara Martin, BCBL Scientific Committee
26/07/23